

- How you design your agents and why?

In this project, we used a pipeline multi-agent collaboration pattern. The system follows a sequential process, where tasks are completed in a fixed order.

The Query Expander takes the research question and breaks it into smaller sub-questions, keywords, and related search queries. These generated queries are then passed to the Source Hunter, which searches the paper corpus using the `search_papers` tool and finds 8–12 relevant passages along with their sources.

Next, the Synthesizer reviews the retrieved evidence and organizes the findings into themes, agreements, disagreements, and research gaps while keeping track of the supporting sources. Finally, the Report Writer uses the synthesized findings to create the final literature review with citations. The final report follows the required markdown structure.

We chose a sequential process because it is well suited for report generation, where each step depends on the previous one. Each agent completes its task before the next agent begins, and the output from one task becomes the input for the next. This creates a clear workflow from research question to final literature review.

- Examples of ReAct-style reasoning in your agent outputs

A ReAct-style example appeared in the Source Hunter agent. The agent used the `search_papers` tool several times to gather evidence for the research question. After each search, it observed the returned passages and continued searching until it had enough relevant evidence. This reflects the ReAct pattern of reasoning, acting through a tool, observing the result, and then deciding the next action.

- What worked, what didn't, and what you would change

What worked well was the sequential design of the crew. Each agent had a clear responsibility, and the output of one agent was passed to the next. The Source Hunter was able to use the `search_paper` tool multiple times and gather relevant evidence, while the Synthesizer and Report Writer successfully organized that evidence into a structured literature review as instructed.

One issue I noticed was that sometimes vector store search failed, and some search results were repetitive or came from the same papers.

To improve the system, I would try to reduce duplicate evidence, and add logic to confirm the passages and citations are accurate and properly matched.

- What would be needed to use this system in production

To use the system in production, it needs some improvements.

- It should keep logs of what each agent did and any errors that happened.
- The results should be validated automatically to make sure the information is accurate and also the citations.
- The system should be tested to make sure it works with more documents without slowing down
- API keys should be stored in vault
- A human-in-the-loop should be there to review the reports for important research.